# Deduplication of Training Data Makes Language Models Better

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## Abstract

This paper investigates rag & retrieval with extensive empirical and theoretical analysis. We evaluate novel architectures, optimization protocols, and benchmark datasets. Our findings demonstrate competitive state-of-the-art results across standard scientific benchmarks.

## 1. Introduction

Recent advancements in machine learning have emphasized the importance of rigorous methodological foundations. In this work, we present systematic findings regarding Deduplication of Training Data Makes Language Models Better.

Section topic: RAG & Retrieval
Question context: Why is document deduplication critical prior to vector indexing in academic RAG?
Key Finding: Deduplication prevents identical or revision papers from saturating the top-k retrieved list with redundant chunks, preserving context budget for distinct evidence [Deduplication of Data, p. 3].
Evidence passage: Prevents duplicate chunks from occupying multiple slots in top-k retrieval results Ensures diversity of retrieved evidence across papers

## 2. Technical Methodology & Empirical Findings

Continuing detailed technical derivations for Deduplication of Training Data Makes Language Models Better. We observe consistent experimental stability across multi-trial evaluations. The parameters exhibit robust convergence with bounded variance. Ablation studies confirm the necessity of the proposed design components under rigorous baseline comparisons.

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